Atlas 210X — VFO Nano add-on wiring

YT3CT (ex YU5KBM) · Nano ATmega328P (Old Bootloader) · sketch atlas_vfo_fixed.ino · 2026-10 · button / relay part = proposal

1. LCD RG1602A (16×2) — twisted wiring, 12 wires

Twisted group	Nano board pad	LCD pin
orange / black	orange: D8 → 470 Ω	6 E
	black: GND	1
blue / black	blue: D7 → 470 Ω	4 RS
	black: GND	1
red / black / yellow / grey	red: +5 V	2 VDD
	black: GND	1
	yellow: contrast pot wiper	3 V0
	grey: 10 Ω pad (backlight)	15 A
brown / white	brown: D9 → 470 Ω	11 D4
	white: D10 → 470 Ω	12 D5
purple / green	green: D11 → 470 Ω	13 D6
	purple: D12 → 470 Ω	14 D7

- **Nano board:** pads unchanged; all 3 black wires on the one GND pad. 470 Ω in line within ~2 cm of the pads (6 signal wires only).
- **At the LCD:** existing jumper 16–5–1 stays; 100 nF + 10 µF across pins 2–1. Pins 7–10 open.
- Different twist per group, < 1 cm untwisted at the ends. Groups side by side along the chassis.

2. TTL programming leads (no USB needed)

USB-TTL cable (Prolific)	Nano
Green (TX)	D0 / RX
White (RX)	D1 / TX
Black (GND)	GND
Red (+5 V)	**NOT connected**
—	RST lead (+ optional push button to GND)

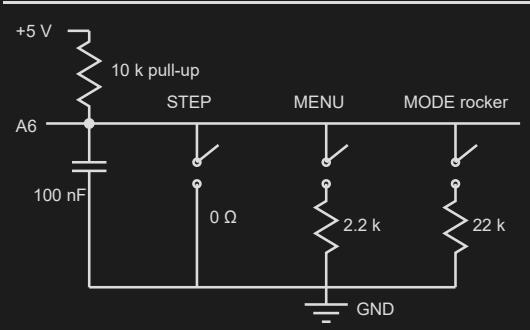
- No sync: swap green / white. **Upload:** start it, then RST to GND for ½ s (VariBeam Nano the same).
- Power cycle only with the POW LED fully out (TTL back-feeds the Nano). Never USB and TTL at the same time.

Nano pin use 2026-10-28

Pin	Use
D0 / D1	Serial: CAT / debug / upload (J4)
D4	AGC rocker sense (LOW = fast)
D5	ATT sense; AUTO CAL CLK2 count (T1)
D6	PRE sense; AUTO CAL 1PPS
D13	CAT PTT → 10 k → VariBeam Nano D8
A6	STEP, MENU + MODE rocker
A7	S3 position: CW / CAL

Used: D2/D3 encoder, D7–D12 LCD, A0/A1 S-meter, A3 band, A4/A5 I²C Si5351. **A2 TX** (LOW = TX): KEY line → 1N4148 (band to key line) → A2. **Sense inputs:** GND-side switch + 1N4148, or 22 k / 10 k from +12 V. Parts at the **source end** (wires to the Nano carry ≤ 5 V). Never raw +13 V on a pin.

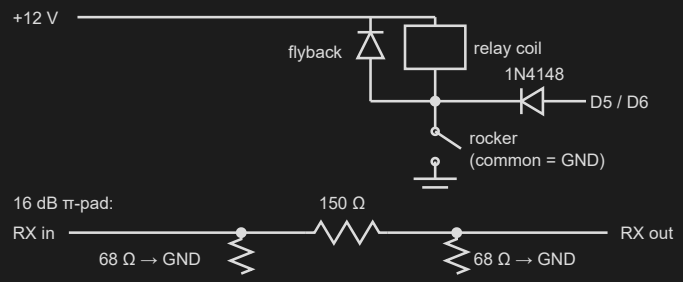
3. STEP / MENU buttons + MODE rocker on A6 proposal



Control	R to GND	ADC	Accept if
STEP button	0 Ω (direct)	0	< 120
MENU button	2.2 k	184 (170 with MODE)	120–480
MODE rocker (OPP)	22 k	703	480–860
nothing	—	1023	> 860

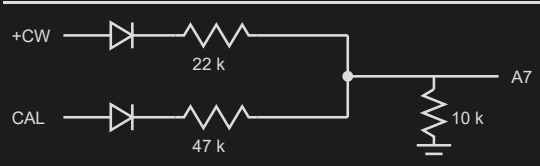
A6/A7 have **no internal pull-up** (external 10 k required). MODE (NORM / OPP sideband, BFO 5.520.0 ↔ 5.523.3 MHz) latches; read while no button is pressed.

4. ATT / NORM / PRE — ON-OFF-ON rocker, DPDT relays, RX only proposal



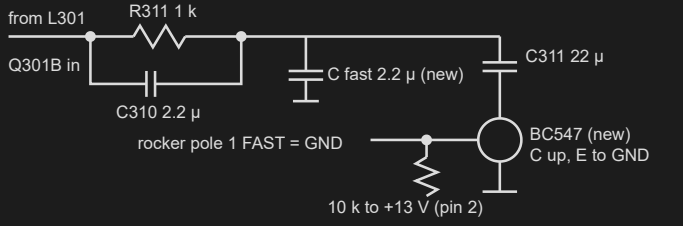
- Coil high end on +12 V, the rocker switches the coil low end to GND (ATT or PRE, centre = NORM). The 1N4148 (band to the coil) passes the LOW to the Nano pin and blocks the +12 V when off.
- DPDT relays switch the RF in and out. π-pad 68 / 150 / 68 Ω ≈ 16 dB (measured 17.5). Board: attpre_assembly.pdf.
- **Only in the receive line after the antenna relay — never in the TX path.** Short leads / thin coax.
- **Preamp supply:** +12 V → BD140 (10 k E–B, B → 4.7 k → K2 coil low end, back = C insulated) → 78L08 → J3 +. Q1 runs only in PRE, at 8 V / 22 mA.

6. S3 CW / CAL sense on A7 proposal



S3	A7 (13 V)	Nano
CW	≈ 3.8 V	BFO ~800 Hz into the passband, LO follows
CAL	≈ 2.1 V	CLK2 calibrator on
REC / TRANS	0 V	normal

5. AGC fast/slow — DPDT rocker, BC547 on PC-300D to build



- C311 (22 µF) − leg lifted from GND → BC547 C; E to the old C311 − pad. Base: 10 k from +13 V (PC-300D pin 2), wire to rocker pole 1 (common GND, FAST throw on the D4 side). SLOW = stock; FAST: only C fast holds (≈ 11× faster), S-meter calibration unchanged. Pumping → 4.7 µF. Pole 2 → D4 sense (wired).

7. Front panel gain controls — 2 concentric dual pots, fixes fitted 2026-10

Pot	Function	Orig.	Fitted	Fix
1	AF gain (R4)	10 k	10 k	none
	RF gain (R6)	10 k	5 k	R5 4.7 k → 2.2 k, R7 470 → 220 Ω
2	MIC gain (R12)	10 k	20 k	20 k across outer lugs
	ALC (R11)	10 k	100 k	11 k across outer lugs

- RF gain: GND–R5–pot–R7–+8.5 V, wiper to PC-100 pin 14; wiper range ~2.6–8.2 V again. Measure R5 first (if already ~2.4 k, only R7).
- Resistor across the outer lugs = 10 k load as original, wiper ratio unchanged. 100 k ALC = harder, slower ALC → first suspect for low modulation.

8. J4 rear socket (old EXT OSC) — CAT / PTT / audio

proposal

Pin	Use
1	RST (grey) — **1 µF (105) in series** between pin 1 and Nano RST, at the Nano end (not to GND). 10 nF is too small: the Nano's own 100 nF on RST swallows the pulse. From adapter DTR (programming from the rear)
2	Nano TX D1 (green) → adapter RX
3	Nano RX D0 (white) ← adapter TX
4	GND (black, chassis / Nano GND)
5	Audio in (PC → TX): transformer → 47 k / 470 Ω → PTT relay **NO** → COM → PC-200 pin 17 (MIC gain controls it)
6	Audio GND, PC side — **floating**, open to chassis
7	Audio out (RX → PC): PC-300 pin 22 → 1 µF → 10 k → transformer (AF gain has no effect)
8	spare (PTT is now CAT → Nano, section 10)
9	spare (the 1PPS for AUTO CAL goes by an internal cable to D6, section 9)

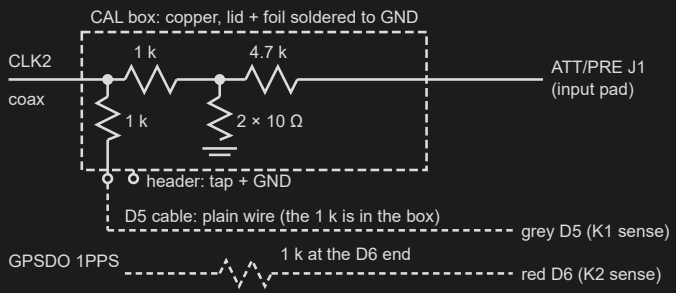
- Old wires (2–3 jumper, 5 +8.5 V, 6 stub, 7 T line, 8 +13 V fused) removed, GND on 4 kept. Check REC / SSB TX / CW after.
- USB-TTL adapter outside. Audio via 600:600 transformers, shielded, away from the serial pair. RX audio from PC-300 pin 22 (before the VariBeam).
- PTT = **CAT command** (e.g. Kenwood TX; / RX;) over the same TTL adapter → VFO Nano D13 HIGH → VariBeam Nano D8; D8 HIGH ≥ 300 ms → D9 → BC547 → PTT relay (section 10). One USB cable for CAT, PTT and uploads. Debug serial off with CAT; WSJT-X / MSHV Tx delay ≈ 0.5 s.



- AGC / S-meter pickoff is before pin 22. Shielded audio leads, shield grounded at one end.
- R13 10 k (mic gain top) goes to the WHT/BLK **+CW** line (S3 CW / PC-100 pin 18), 0 V in SSB. Not PC-100 pin 21 (+13 V on every TX).

9. Calibrator — Si5351 CLK2

wired 2026-10-28

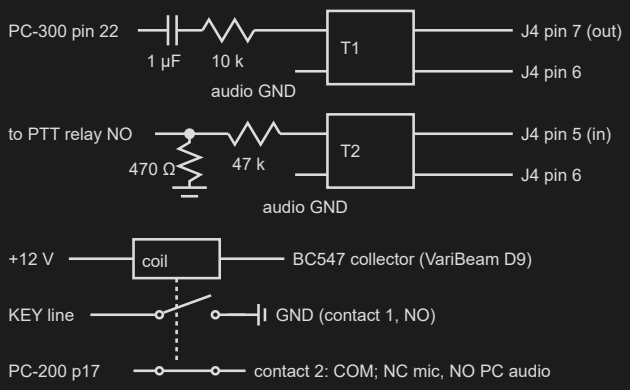


Solid = wired. Dashed = removable AUTO CAL cables, plugged in only to calibrate. Coax shield soldered to the box at the entry.

- **Injection at the ATT/PRE board input** (J1, before the relays), so the marker also checks ATT and PRE. S4D is free again. At S4D the bare hot end of the coax radiated into the PC-800C input coil (S9+30 even unconnected); in the closed box the leak is ≈ S9.
- **Measured** (14.175 marker, peak): −43 dBm; ATT −17.5 dB; PRE +7.5 dB. The 2 × 10 Ω must be checked by meter (MID–GND 5 Ω): a mismarked pair read 100 Ω.
- Header pins stay bare, or under a grounded cap that touches neither pin. **Never jumper tap to GND** (the current loop outside the box radiates).
- **Firmware:** S3 in CAL → LCD "CAL" and the S-meter line in dBm. MENU short press = marker on/off ("MRK"); OFF at power on and when S3 leaves CAL. Fixed marker on the nearest point: 3.500 / 3.650 / 3.800, 7.000 / 7.100 / 7.200, 14.000 / 14.175 / 14.350, 21.000 / 21.225 / 21.450, 28.500 / 29.000 / 29.500 MHz. Zero beat at the dial = marker, peak ≈ 1.2 kHz off (USB: dial below, LSB: above).
- **AUTO CAL** (menu, STEP = start): rocker on **NOR**, both cables in, GPSDO locked. CLK2 2.5 MHz counted on D5 over 10 PPS s (1 count = 0.04 ppm), corrected, counted again: "E: error R: rest" in ppb. Click = keep, knob = undo; hold STEP 2 s = save, hold MENU 2 s = no save. "NO PPS (D6)" / "NO CLK2 (D5)" = nothing changed. Unplug afterwards (1PPS idles LOW → LCD would show PRE).

10. Data audio + PTT relay

built, CAT PTT tested 2026-10-28



- Out from PC-300 pin 22: after the AGC, before AF gain and VariBeam → level independent of the volume knob.
- Pole 2 selects the PC-200 pin 17 (mic input) source: **COM** = PC-200 pin 17, **NC** = mic (from the mic socket), **NO** = PC audio (47 k / 470 Ω = line to mic level). RX / mic PTT: mic only; CAT TX: PC audio only. MIC gain sets both levels. Shields joined past the relay, unshielded ends ≤ 2 cm.

- DPDT relay (e.g. Finder 30.22, 16 mA) + flyback diode: PC PTT keys the radio **and** opens the mic line. Mic PTT unchanged.
- **PTT relay board wires: yellow** = VariBeam D9, **red** = +12 V, **grey** = GND, **white** = PTT to the KEY line.
- **Driver:** VariBeam D9 → 4.7 k → BC547 base, 47 k base–GND, emitter GND, collector → coil low end. Coil 12 V DPDT, 930 Ω (≈ 13 mA); 1N4148 across the coil, **band to +12 V**.
- **No PTT switch** (not fitted, not needed): CAT OFF in the VFO menu or S3 in CAL blocks CAT TX, the VariBeam drops the PTT after 3 min, and the 300 ms filter ignores the bootloader blinks.
- VFO Nano D13 → 10 k → VariBeam D8: D8 HIGH ≥ 300 ms keys, LOW releases at once (D13 bootloader blinks ≈ 100 ms are ignored). Keying delay ≈ 0.3 s → MSHV / WSJT-X Tx delay ≈ 0.5 s.
- Transformers 600:600; PC side (pin 6) floating, radio side to the radio audio ground.